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MILITARY STANDARD

GLOSSARY OF TERMS AND DEFINITIONS FOR ULTRASONIC TESTING PROCEDURES

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

MIL-STD-371
Glossary of Terms and Definitions For Ultrasonic Testing Procedures

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FOREWORD

This standard prescribes terminology for use with ultrasonic test methods and procedures.

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1. SCOPE

1.1 Purpose. The terminology in this standard for ultrasonic test methods and procedures is intended for use on drawings, and in specifications, standards, and technical documents.

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2. REFERENCED DOCUMENTS

2.1 Government documents. Not applicable.

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3. ULTRASONIC TERMS AND DEFINITIONS

AMPLITUDE The vertical pulse height of a received ultrasonic signal, usually base to peak, indicated by an A-scan presentation.

ANGLE BEAM A wave train introduced into a test object so that the ultrasonic beam axis centerline is at an angle, other than 0 degrees, to the normal to the entry plane of the test object.

ANGLE OF INCIDENCE The angle that the axis of an ultrasonic beam makes with the normal to a surface at the point of incidence.

ANGLE OF REFLECTION The angle that the axis of a reflected ultrasonic beam makes with the normal to a reflecting surface at the point of incidence.

ANGLE OF REFRACTION The angle between the axis of a refracted beam and the line normal to the boundary between two media with different speeds of sound.

AREA-AMPLITUDE RESPONSE CURVE A curve showing the changes in the amplitude of ultrasonic response to reflectors of different areas located at equal distances from the search unit in an ultrasonic-conducting medium.

A-SCAN A method of data presentation utilizing a horizontal base line that indicates distance, or time, and a vertical deflection from the base line which indicates ultrasonic amplitude.

ATTENUATION The loss of ultrasonic energy within a medium due to scattering and absorption.

ATTENUATION COEFFICIENT Factor that describes the decrease in ultrasound amplitude with distance in a given medium. Normally expressed in decibels (dB) per unit length.

ATTENUATOR A device for altering the amplitude of an ultrasonic signal in known increments, usually decibels.

BACK REFLECTION Indication of the ultrasonic echo from the far boundary of the material under test.

BANDWIDTH A measure of the range of frequencies over which the output of a device (e.g., transducer, pulser, or amplifier) or system contains the most energy.

BASELINE The distance/time axis in an A-scan display.

BEAM AXIS The acoustic centerline of an ultrasonic search unit's beam pattern as described by the locus of points of maximum sound pressure in the far field, and its extension into the near field.

BEAM SPREAD A divergence of the ultrasonic wave train as it travels through a medium.

BOTTOM ECHO or BACK ECHO See Back Reflection.

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BROADBAND A descriptor for a device (e.g., transducer, pulser, or amplifier) or system with a relatively wide bandwidth.

B-SCAN PRESENTATION A method of data presentation in which the travel time of an ultrasonic pulse is represented as a displacement along one axis, and probe movement (generally rectilinear) is represented as a displacement along the other axis. The display presents a cross-section of the test piece and reflected pulses are shown in contrast to the background.

BUBBLER A device using a liquid stream to couple an ultrasonic beam to the test piece.

COMPRESSIONAL WAVE See Longitudinal Wave.

CONTACT TESTING A method of testing in which the transducer contacts the test surface, usually through a thin layer of couplant.

CORNER EFFECT The reflection of a sound beam in a direction opposite and parallel to an incident beam which has been directed normal to the intersection of two perpendicular planes. The reflecting corner being two intersecting surfaces at right angles to each other.

COUPLANT A substance used between a search unit and test surface to permit or improve transmission of ultrasonic energy.

CRITICAL ANGLE The incident angle of a sound beam beyond which a specific refracted or reflected mode of vibration can no longer exist.

CROSS TALK Signal leakage (acoustic or electrical) across an intended barrier.

CRT Abbreviation for cathode ray tube.

CRYSTAL See Element, Piezoelectric.

C-SCAN A method of data presentation yielding a plan view of the test object and detected discontinuities.

DAMPING Limiting the duration of vibration in an ultrasonic search unit by either electrical or mechanical means.

DEAD ZONE Corresponds to the distance from the surface of the test object to the nearest inspectable depth. It is determined by the characteristics of the material, ultrasonic test instrument and transducer.

DECIBEL (dB) Twenty times the base ten logarithm of the ratio of two ultrasonic signal amplitudes. $dB = 20 \log_{10}$ (amplitude ratio).

DEFECT A discontinuity or group of discontinuities which produce indications that meet or exceed a specified reject criteria.

DELAY LINE Material (liquid or solid) placed in front of the search unit designed to introduce a time delay between the initial pulse and reflected pulses.

DELAYED SWEEP A horizontal sweep whose start is delayed in order to prevent the appearance of unwanted early response information on the display.

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DGS-DISTANCE GAIN SIZE Distance amplitude curves permitting prediction of reflector size compared to the response from a back surface reflection.

DISCONTINUITY A detectable interruption in the material which may or may not have undesirable connotations.

DISTANCE AMPLITUDE CORRECTION (DAC) (Swept Gain, time corrected gain, time variable gain, etc.) Electronic change of amplification necessary to provide equal amplitude from equal reflectors at different distances.

DISTANCE-AMPLITUDE RESPONSE CURVE A curve relating ultrasonic echo amplitudes from equal reflectors at different distances in the material.

DIVERGENCE See Beam Spread.

DUAL SEARCH UNIT (TWIN PROBE) A search unit containing two elements, (side-by-side) one a transmitter, the other a receiver .

DYNAMIC RANGE The ratio of maximum to minimum reflective areas that can be distinguished on the CRT at a constant gain setting.

ECHO Indication of reflected energy.

ELEMENT, PIEZOELECTRIC Portion of a single crystal or polycrystalline sintered ceramic having piezoelectric properties, used for the generation and/or detection of ultrasonic energy.

EQUIVALENT FLAT BOTTOM HOLE A flat bottomed hole reflector in a similar material and geometry with a diameter that produces the same ultrasonic echo amplitude as the reflector under evaluation.

EXPANDED SWEEP An enlargement of the time base of the signal display.

FALSE INDICATION In nondestructive inspection, an indication that may be misinterpreted as a discontinuity or defect; a non-relevant indication, e.g., artifacts.

FAR FIELD The zone of the ultrasonic beam of a non-focused search unit that extends beyond approximately $D^2/4\lambda$ (where D is the diameter of the transducer and λ is the wavelength). In this zone the amplitude of the ultrasonic waves decrease monotonically with distance.

FLAW A discontinuity in the material which is generally undesirable, but may or may not be severe enough to immediately render the part unfit for intended use.

FOCUSED BEAM Converging energy of the sound beam at a specified distance.

FOCUSED SEARCH UNIT A search unit which produces a focused beam traditionally by means of an acoustic lens.

FREQUENCY (ACOUSTIC) The number of oscillations of cycles per second (expressed in Hertz (Hz)) experienced by a particle or point in a medium caused by the passage of an acoustic wave through it.

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FREQUENCY (CENTER) See Frequency (Dominant).

FREQUENCY (DOMINANT) That frequency at which the overall response of an ultrasonic pulse-echo flaw detection system is a maximum. NOTE: the 'system' includes the pulser, the transducer as a transmitting element, the pulse propagation path, the transducer as a receiving element, and electronic instrumentation associated with receiving, amplifying and displaying a received signal.

FREQUENCY (EXAMINATION) The nominal frequency at which the examination is performed.

FREQUENCY (FUNDAMENTAL RESONANCE) The lowest acoustic frequency which will cause a condition of resonance to be established in a given material of given thickness. NOTE: This frequency will be such that the wavelength in the given material is equal to twice the given thickness.

FREQUENCY (PULSE REPETITION) The number of times per second an electroacoustic transducer is excited to produce a pulse of acoustic energy.

FRESNEL FIELD (FRESNEL ZONE) See Near Field.

GATE An electronic means to select a segment of the sweep range for monitoring or further processing of ultrasonic signals.

GATE THRESHOLD An adjustable electronic signal level such that while any ultrasonic signal within the gate exceeds the set level, an on/off signal (e.g., to a light or a recording pen) is activated.

GHOST ECHO See Wrap Around.

GRASS See Hash.

GRAZING INCIDENCE Immersion inspection with the beam directed at a glancing angle to the test surface.

HASH Numerous small indications along the baseline of the display indicative of background noise. Hash is sometimes caused by many small inhomogeneities in the material.

HOLOGRAPHY (ACOUSTIC) An inspection system using the phase interference between ultrasonic waves from an object and a reference signal to obtain an image of reflectors in the object under test.

IMMERSION TESTING An examination method where the transducer and the test material are submerged (at least locally)in a liquid, usually water.

IMPEDANCE (ACOUSTIC) The product of density and sound speed for a given material. The property which determines acoustic transmission/reflection characteristics at a boundary between two media.

INDICATION In nondestructive evaluation, evidence of a discontinuity that requires interpretation to determine its significance.

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INITIAL PULSE (MAIN BANG) Response of the ultrasonic system display to the transmitted pulse.

INTERFACE The boundary between two materials.

IRRELEVANT INDICATION An indication other than one from a discontinuity of interest.

LAMB WAVE A type of wave that propagates within the thickness of a plate, and that can only be generated at particular values of angle of incidence, frequency, and plate thickness. The velocity of the wave is dependent on the mode of propagation and the product of plate thickness and frequency.

LINEARITY (AMPLITUDE) The ability of an ultrasonic testing system to display amplitudes proportional to amplitudes of ultrasonic waves coming back from a reference reflector.

LINEARITY (DISTANCE) The ability of an ultrasonic testing system to display signals at spacings proportional to the sound path distance between corresponding reflectors.

LONGITUDINAL WAVE Those waves in which the particle motion of the material is in the same direction as the wave propagation.

LOSS OF BACK REFLECTION Absence of or a significant reduction of an ultrasonic echo from the back surface of the article being inspected.

MANIPULATOR A device, generally used in immersion techniques, to control the orientation of the search unit.

MATERIAL ENVELOPE The portion of material at the surface of a test piece which will later be removed to produce the finished part.

MODE The manner in which acoustic energy is propagated through a material as characterized by the particle motion of the ultrasonic wave.

MODE CONVERSION The process by which a wave of a given mode of propagation generates waves of other modes of propagation by reflection or refraction.

MULTIPLE BACK REFLECTIONS Successive echoes from the far boundary of the material being examined.

NEAR FIELD The zone directly in front of an unfocused ultrasonic transducer extending to a distance of approximately $D^2/4\lambda$ (where D is the diameter of the transducer and λ is the wavelength). In this zone components of the ultrasonic wave from different portions of the transducer reinforce and cancel to produce pressure maxima and minima.

NOISE Unwanted disturbances from equipment or the material under test superposed upon the received ultrasonic signal.

NORMAL INCIDENCE Condition in which the axis of the transducer beam is perpendicular to the test surface.

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PAINTBRUSH TRANSDUCER A rectanglar transducer usually constructed with a number of piezoelectric elements and has a wide beam to sweep out large areas.

PENETRATION The maximum depth in a material from which useful ultrasonic back reflections can be obtained.

PIEZOELECTRIC EFFECT The characteristic of certain materials to generate electrical charges when subjected to mechanical vibrations, and conversely to generate mechanical vibrations when subjected to electrical pulses.

PITCH-CATCH An ultrasonic technique in which the energy is emitted by one transducer and received by another on the same or adjacent surface.

PLATE WAVE See Lamb Wave.

PROBE See search unit or transducer.

PULSE A transient ultrasonic or electrical signal.

PULSE ECHO METHOD An ultrasonic inspection method in which the presence and position of a reflector are indicated by the reflected pulse amplitude and time and where one search unit serves as both generator and receiver of acoustic energy.

PULSE LENGTH A measure of the duration of a wave train, expressed in time or number of cycles.

PULSE RATE See Frequency (Pulse Repetition).

RF (RADIO FREQUENCY) DISPLAY A display showing unrectified ultrasonic signals.

RANGE The ultrasonic path length that is displayed.

RAYLEIGH WAVE An ultrasonic surface wave in which the particle motion is elliptical and the effective penetration is approximately a wave length. The waves follow the curvature of the part and reflections occur only at sudden changes in the surface.

REFERENCE BLOCK A specimen with geometry and material designed to produce ultrasonic reflections of known characteristics for purposes of comparison.

REFLECTION See Echo.

REFLECTOR An interface at which a ultrasonic beam encounters a change in acoustic impedance, and reflects at least part of its energy.

REFRACTION The change in direction of the ultrasonic beam as it passes obliquely from one medium to another medium, with a different sound wave velocity.

REJECT (SUPPRESSION) A electronic control which minimizes or eliminates low amplitude signals (or noise) but may cause display non-linearity.

RESOLUTION The ability of ultrasonic equipment to separate indications from reflectors that are located closely adjacent to one another.

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RESONANCE METHOD A technique in which continuous ultrasonic waves are varied in frequency to discriminate some property of the part, such as thickness stiffness, or bond integrity.

SATURATION Condition observed on the display resulting from a signal of such a magnitude that an increase in the signal produces no observable increase in the display amplitude.

SCANNING Relative movement of the transducer and the test piece in order to interrogate a volume of material.

SCATTERING The perturbation of a ultrasonic beam by small reflectors in the propagation path.

SCANNING INDEX The distance the transducer is moved perpendicular to the traverse direction after one traverse of the part.

SE PROBE See Dual Search Unit.

SEARCH UNIT OR TRANSDUCER An electro-acoustic device used to transmit and/or receive ultrasonic energy in order to interrogate a specimen. The device frequently consists of piezoelectric element(s), backing material, case, connector and a front protective covering or lens or wedge.

SENSITIVITY A measure of the ability of an ultrasonic system to detect small discontinuities.

SHEAR WAVE Wave motion in which the particle motion is perpendicular to the direction of ultrasonic propagation.

SHEAR WAVE TRANSDUCER A transducer used for generating and detecting normal incidence shear waves; e.g., y-cut quartz transducer.

SHOE See WEDGE

SIGNAL-TO-NOISE RATIO The ratio of the amplitude of an ultrasonic indication to the amplitude of the background noise.

SKIP DISTANCE In angle beam testing, the distance along the test surface from the sound entry point to the point at which the sound returns to the same surface, having been reflected from the far surface of the test object.

STRAIGHT BEAM See Normal Incidence.

SURFACE WAVE See Rayleigh Wave.

SWEPT GAIN See distance amplitude correction (DAC).

THROUGH TRANSMISSION METHOD A method in which reflectors in an object are detected by monitoring the reduction in ultrasonic energy incident on a receiving transducer after the energy has propagated through the object from a transmitting transducer.

TRANSDUCER See Search Unit.

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TRANSFER MECHANISM A procedure to account for differences in ultrasonic response due to differences in surface texture, curvature, attenuation, etc., between a reference block and the test object.

TRANSVERSE WAVE See Shear Wave.

ULTRASONIC Pertaining to mechanical vibrations having a frequency greater than approximately 20,000 Hz.

ULTRASONIC SPECTROSCOPY Analysis of the frequency content of an ultrasonic wave.

VEE PATH The angle-beam path in material starting at the transducer examination surface, going through the material to the back surface, continuing to the examination surface in front of the search unit, and reflecting back along the same path to the transducer if a discontinuity is encountered. The path is usually shaped like the letter "V". See also Skip Distance.

VERTICAL LIMIT The maximum readable level of vertical indications determined either by an electrical or a physical limit from an A-scan presentation.

VIDEO PRESENTATION Display of the rectified and usually filtered rf ultrasonic signal.

WATER PATH (WATER TRAVEL) The distance from the transducer to the test surface in immersion or water column testing.

WAVE FRONT A continuous surface drawn through the most forward points in a wave disturbance which have the same phase.

WAVE INTERFERENCE A series of pressure minima and maxima resulting from the superposition of waves, resulting in wave cancellation and reinforcement.

WAVELENGTH The distance in the medium in the direction of propagation for a wave to go through one complete cycle. The wavelength λ is given by $\lambda = c/f$ where c is the sound speed in the medium and f is the frequency (acoustic) of the wave.

WEAR FACE Material attached to the face of a search unit to prevent wear of the transducer element.

WEDGE An angle block, usually of plastic material, placed in acoustic contact between an ultrasonic transducer and the test object which causes ultrasonic waves to propagate at a predicted angle into the test object.

WHEEL TRANSDUCER An ultrasonic device incorporating one or more piezoelectric elements mounted inside a liquid filled flexible tire. The beam is coupled to the test surface through the rolling contact area of the tire.

WRAP AROUND Display of misleading ultrasonic reflections from a previously transmitted pulse due to the use of excessive pulse repetition frequency.

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4. NOTES

4.1 Subject term (key word) listing.

Glossary of NDTI terms
Nondestructive Testing
Ultrasonic Testing

Custodians:
Army -- MR
Navy -- AS
Air Force -- 20

Review activities:
Army -- AR, AV, EA, MI
Navy -- SH
Air Force -- 82, 99

User activities:
Army -- AL, AT, NE, TE, WV
NAVY -- OS
Air Force -- 20, 99

Preparing activity:
Army - MR

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